Fiscal Year:	FY 2019	**Task Last Updated:**	FY 06/14/2019
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Spectral Imaging within The EDEN ISS Project – An Antarctic Analog for Enhancing Exploration Life Support		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Space Biology Unsolicited
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/14/2020 per NSSC information (Ed., 11/13/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa Ph.D. (Co PI: University of Florida)		
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Performance Goal No.:			
Performance Goal Text:			
	This project brings established research from NASA funded science on the International Space Station (ISS) into an international project led by DLR (German Space Agency) toward the functionalization of plant growth in support of human exploration. Imaging technologies developed as part of ongoing ISS science will be deployed within the Antarctic EDEN ISS plant production unit to provide tele-science support and plant science analytics to better understand the responses of plants, and production support to optimize plant performance. Plant production for advanced life support within deep space and extraterrestrial colonies has long been an appreciated area of technology development. Indeed ISS science is now turning toward small scale plant production in space, and plant production units in extreme environments on Earth have proven the value of fresh food. However, there has as yet to be any integrated tests of plant production within space vehicle-like structures in extreme analog environments. To		

Task Description:

address this gap in knowledge, the EDEN ISS project by the DLR German space agency is designed to place an ISS-inspired module in the context of an established Antarctic station, drawing together for the first time structures and production units designed to space vehicle formats together with the challenges of an extreme and isolated environment. In keeping with the spirit and intent of the EDEN ISS project, the current proposal seeks to fund US participation, bringing plant imaging technologies developed, in part from extensive Shuttle and ISS plant growth experiments, into the science to be conducted within EDEN ISS. The area of fidelity to be addressed by this project is a series of science investigations designed to further planetary research in terrestrial extreme environments that may be analogous to those found on other planets.

The EDEN ISS project is constructing essential controlled environment agriculture technologies using both an International Standard Payload Rack cultivation system to relate to on-board ISS and cis-lunar deployments for crew food supplementation, and a Future Exploration Greenhouse that is designed with respect to future planetary bioregenerative life support system deployments. The overall physical facility, referred to as the EDEN ISS Mobile Test Facility, will provide realistic mass flow and will deploy to the highly-isolated Antarctic Neumayer Station III, operated by the Alfred Wegener Institute.

Among the technologies to be developed and deployed will be imagers designed by the University of Florida (UF) Principal Investigator (PI) team, through this proposed project, to monitor plant health within the Mobile Test Facility. Our major goal is to equip the plant growth systems with information-rich imaging such that plant health and basic science data can be reviewed by a large and integrated science backroom back on “Earth.” An additional goal is to optimize the conceptual connections between EDEN ISS and current ISS capabilities and operations. These two goals bring a directed focus to the project, connecting the planetary surface analog concepts to near term operations of ISS and potential cis-lunar habitats.

To accomplish these goals, spectral imaging of plants will be incorporated into typical data monitoring systems and operation of the project. EDEN ISS, as a large multinational project, has an extensive Data Management Plan (DMP). The work proposed here will depend and draw upon the EDEN ISS DMP for the overall project data, and will specifically draw Mobile Test Facility (MTF) plant growth images and operations images as primary data for analysis and management by the UF team. The UF team will archive and make available all plant health images specific to our tasks via a separate project within the NASA Life Sciences Data Archive (LSDA). The technology is based on small, robust cameras that image wavelengths related to previous and future ISS hardware such that data from EDEN ISS will provide feed forward science for known and anticipated ISS and habitat plant growth hardware.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Exploration mission scenarios are increasingly engaging plant production as adjunct or major parts of life support. This is especially true for deep space and extraterrestrial surface scenarios, and there have been several lower fidelity projects that include plant production in analog tests of such scenarios. Plant growth has been a part of some simulations, and small-scale production type plant growth units are currently on the ISS. However, in none of these cases has an exploration-realistic plant growth facility been incorporated into a challenging analog site such as Antarctica to move the technology significantly toward anticipated mission structures and operations. And none incorporated sophisticated plant imaging into remote science backrooms. The EDEN ISS project is envisioned to fill that large gap, placing an ISS-inspired module near the German Antarctic Neumayer Station III. The EDEN ISS project is an active, funded project that involves a wide international team and is led by the German Aerospace Center, DLR. We have been invited to contribute our presence, expertise, and imaging technology to EDEN ISS. Opportunities for non-European participation at Neumayer III are rare, and EDEN ISS is unique in its magnitude and fidelity. This proposal seeks to seize the opportunity for important imaging technology advancement and US participation within the multinational context of the EDEN ISS project.
The EDEN ISS project seeks to create a structurally and operationally realistic analog to a space station, vehicular, or planetary habitat plant growth module for life support. As such, the project falls under the purview of Space Exploration and Spaceport Technology Development (SESTD) in several subcategories: In-Situ Resource Utilization Technologies (in that EDEN is dedicated to local production of food for bioregenerative Advanced Life Support), Lunar and Mars Base/Settlement Technologies (in that we will be within a robust simulation of life support technologies), and also Advanced Sensors and Detectors (in that we will be developing imaging cameras for monitoring plant health within the facility). As a working production facility, EDEN ISS will provide fresh produce to Neumayer III overwintering crews. And in the process, the project will validate key plant growth system technologies and operational protocols for use on the ISS and within future planetary bioregenerative life support systems -- a key set of technologies for long duration spaceflight and planetary habitation.

The project also has the potential to substantially contribute to plant health monitoring technology in terrestrial environments. Protected agriculture (production greenhouses and other crop facilities that are environmentally controlled or where climate is ameliorated) is a rapidly growing field in a world with expanding needs for food, and expanding regions of marginal crop land. Robotic tending and monitoring approaches are being embraced as time and resources saving tactics, and the type of health monitoring we are employing at EDEN ISS will contribute to the sophisticated tool box being employed in this expanding technology.

2019 Investigator Expedition
Dr. Robert Ferl and Dr. Anna-Lisa Paul traveled to Neumeyer III Antarctica Research Station in January 2019 for a four week expedition departing from Cape Town, South Africa. While onsite, PIs updated GoPro Imagers and modified the HikVision cameras to add to the filtered imaging options available to monitor plant health in the FEG (Future Exploration Greenhouse).

2019 EDEN Imaging

In 2019 PI’s recalibrated some of the original GoPro cameras and also modified and installed six new cameras based on the Hikvision network cameras being used as part of the other imaging system in the FEG. Modifications were done at Neumeyer III.

Three different filters were used:

Task Progress:	1. NDVI-7 – the original NDVI filter. These are thin and small enough to fit in-frame with the existing housing, and could be set in place with a dot of super glue 2. New Dual Band Pass filters (DBP) – These are round, thick glass with optical coatings. These round filters were about 1 mm too wide to fit perfectly into the Hikvision camera framework, so one edge was ground down with a steel file to enable a flush fit. Filer was secured with dots of super glue on edges 3. Blue film (BF) – an economical option suitable for educational outreach applications; schools could easily purchase and adapt these filters, also easily cut to fit with scissors. After the filters were secured and glue dry, the cameras were individually focused before the cameras were reassembled into their housing. The UF cameras were installed in seven different locations throughout the FEG to monitor a variety of crops. The Hikvision deployment of the original NDVI-7 filter is being compared to the new DBP filter in Hikvision cameras by using both to observe the same tray of plants (bush tomatoes in L1). The single remaining GoPro NDVI-7 camera was deployed to monitor a set of NASA-relevant crops in R1. Veggie Camera Development Early work has been done to develop a plant health monitoring imager for the Veggie hardware aboard the ISS. A new camera that fits the needs and requirements of the hardware has been selected, Brinno TLC200 Pro and has been fitted with the IR corrected lens and necessary adapters. The Brinno camera can operate for approximately 1 and a half months, taking 1-2 images a day, on 4 AA batteries. Veggie imaging SVT will take place during the Veg04/05 experiment. Due to the light spectrum applied in Veggie, filter selection is still ongoing. However, early testing in Veggie has shown multiple possibilities of filters to be used. During the SVT experiment, we will finalize filter selection and determine hardware mount for the imager.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Beisel NS, Callaham JB, Sng NJ, Taylor DJ, Paul A-L, Ferl RJ. "Utilization of single-image normalized difference vegetation index (SI-NDVI) for early plant stress detection." Applications in Plant Science. 2018 Oct 19;6(10):e01186. https://doi.org/10.1002/aps3.1186 ; PMID: 30386712 ; PubMed Central PMCID: PMC6201722 , Oct-2018